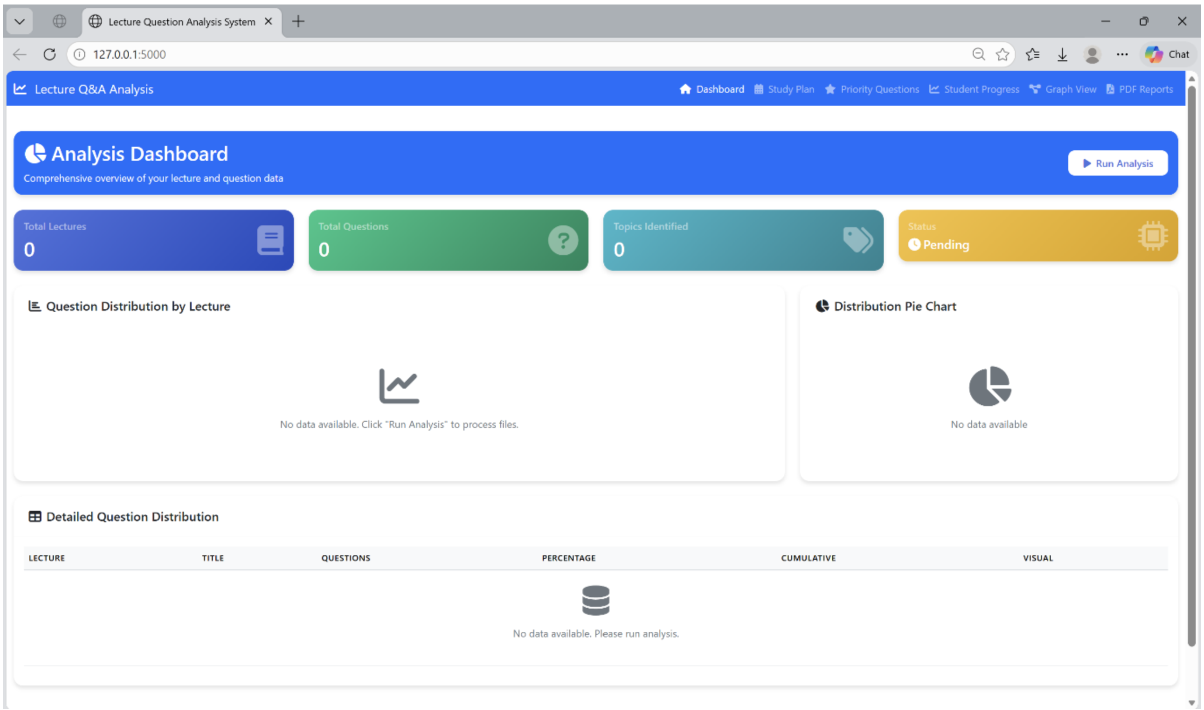
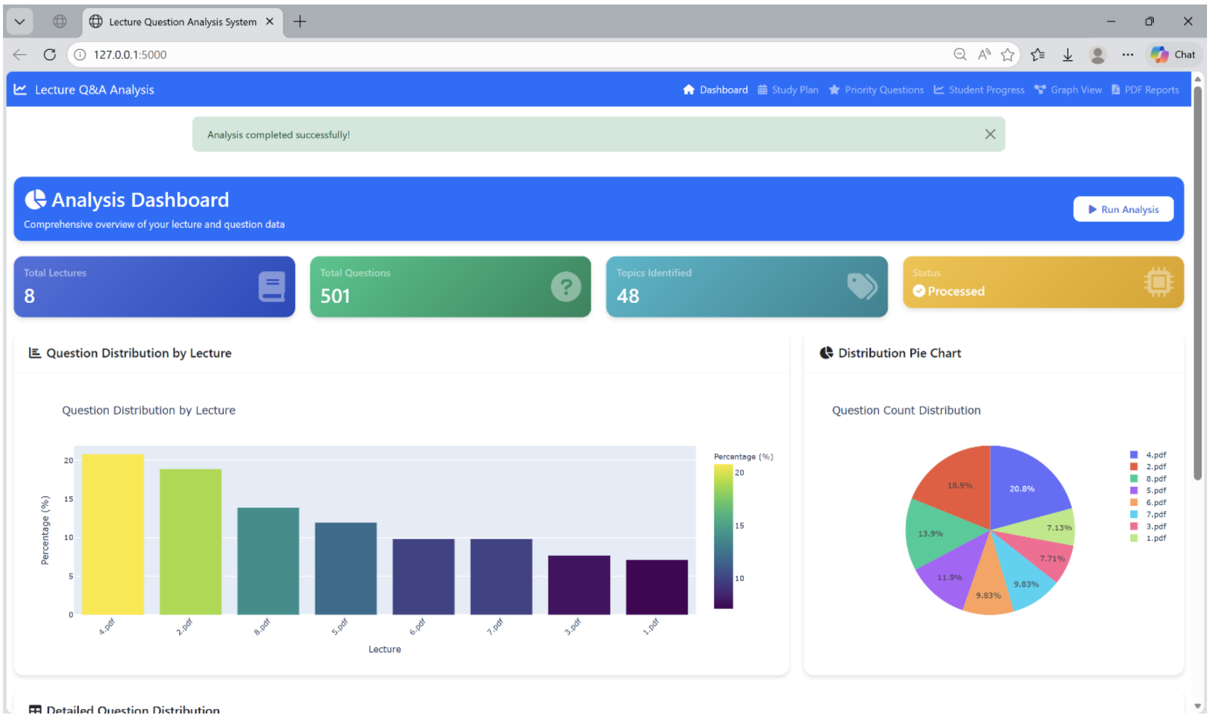
Student Study Plan Generation

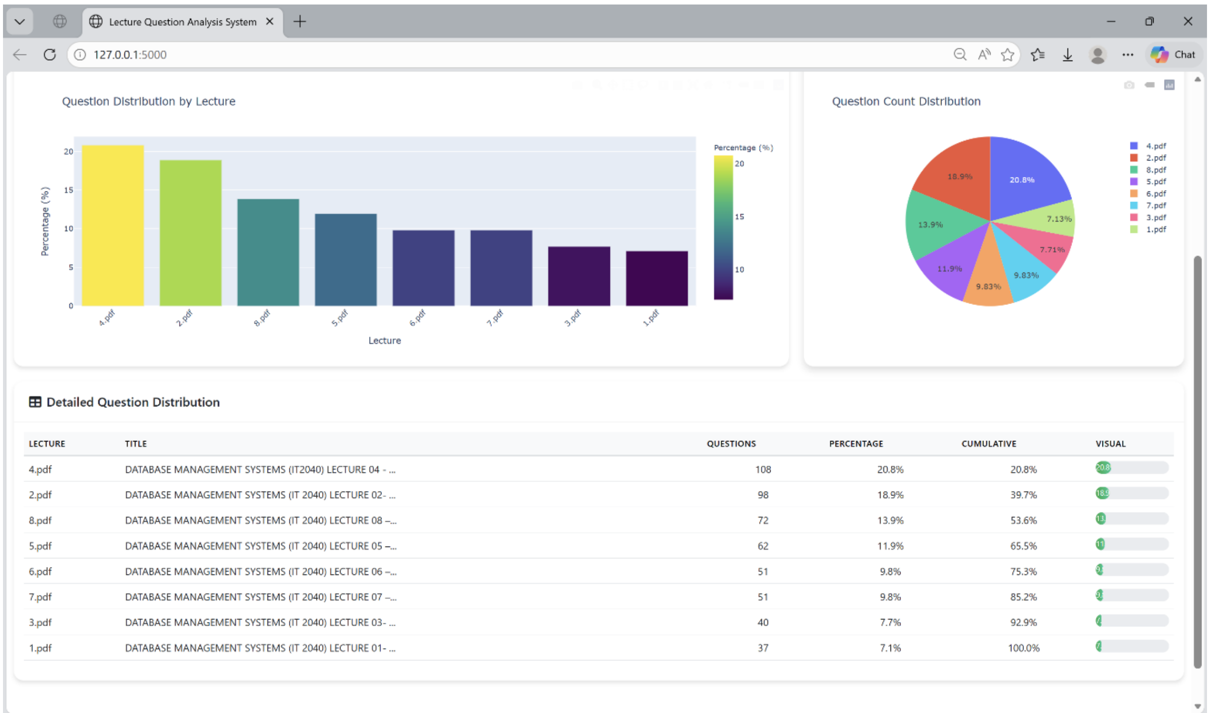
1. Analysis Dashboard

This page shows the main analysis dashboard and lecture-wise question distribution. Initially, the dashboard displays zero values before the analysis is executed. After clicking the **Run Analysis** button, the system processes DMS lecture slides and past MID paper MCQ questions. It extracts questions, identifies lecture topics using Natural Language Processing techniques, and determines the total number of lectures and questions.

After the analysis is completed, the system visualizes the results using charts and statistics. The bar chart and pie chart show how frequently questions from each lecture appear in past exams, helping identify high-priority lectures. The detailed table also provides lecture-wise statistics such as the number of questions and percentage distribution.



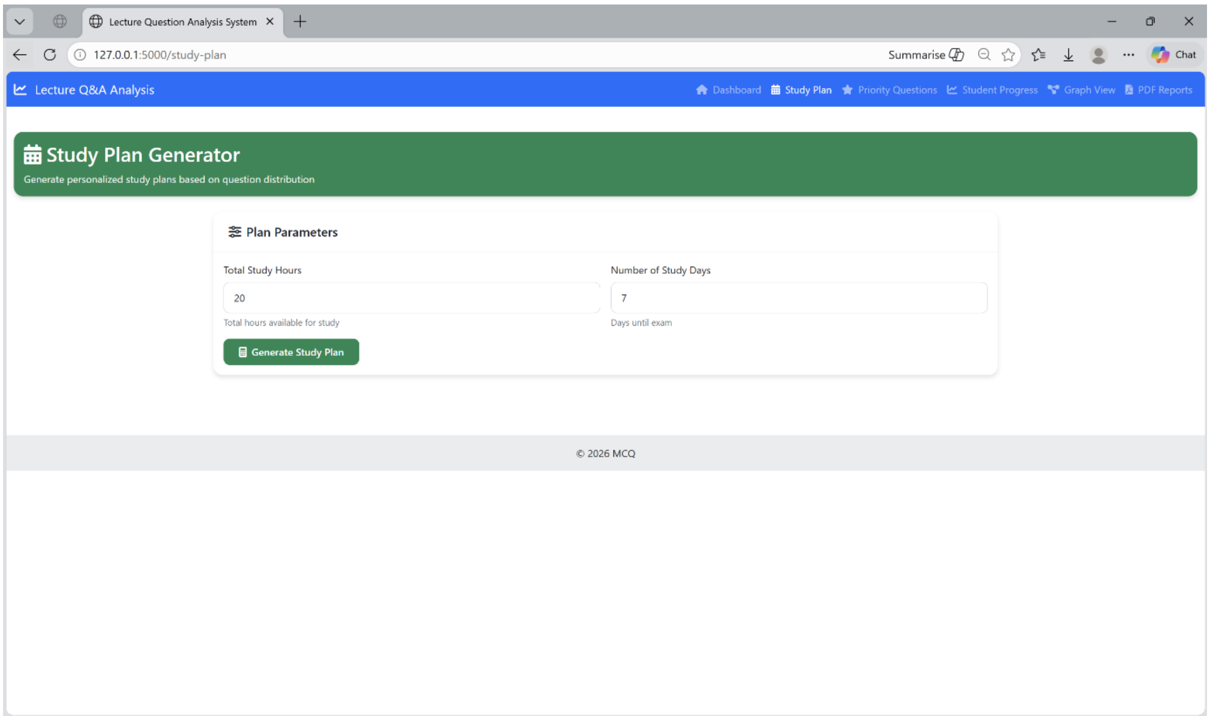


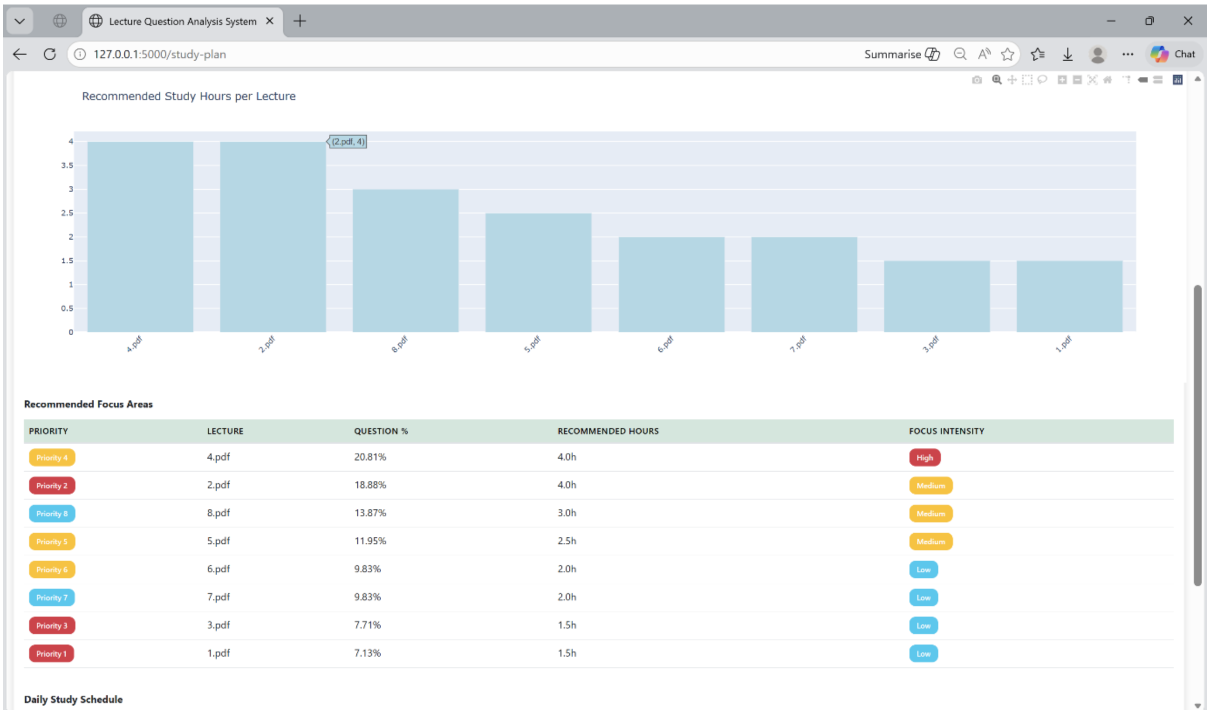


LECTURE	TITLE	QUESTIONS	PERCENTAGE	CUMULATIVE	VISUAL
4.pdf	DATABASE MANAGEMENT SYSTEMS (IT2040) LECTURE 04 - ...	108	20.8%	20.8%	20.8
2.pdf	DATABASE MANAGEMENT SYSTEMS (IT 2040) LECTURE 02- ...	98	18.9%	39.7%	18.9
8.pdf	DATABASE MANAGEMENT SYSTEMS (IT 2040) LECTURE 08 –...	72	13.9%	53.6%	
5.pdf	DATABASE MANAGEMENT SYSTEMS (IT 2040) LECTURE 05 –...	62	11.9%	65.5%	
6.pdf	DATABASE MANAGEMENT SYSTEMS (IT 2040) LECTURE 06 –...	51	9.8%	75.3%	
7.pdf	DATABASE MANAGEMENT SYSTEMS (IT 2040) LECTURE 07 –...	51	9.8%	85.2%	
3.pdf	DATABASE MANAGEMENT SYSTEMS (IT 2040) LECTURE 03- ...	40	7.7%	92.9%	
1.pdf	DATABASE MANAGEMENT SYSTEMS (IT 2040) LECTURE 01- ...	37	7.1%	100.0%	

2. Baseline Study Plan Generation

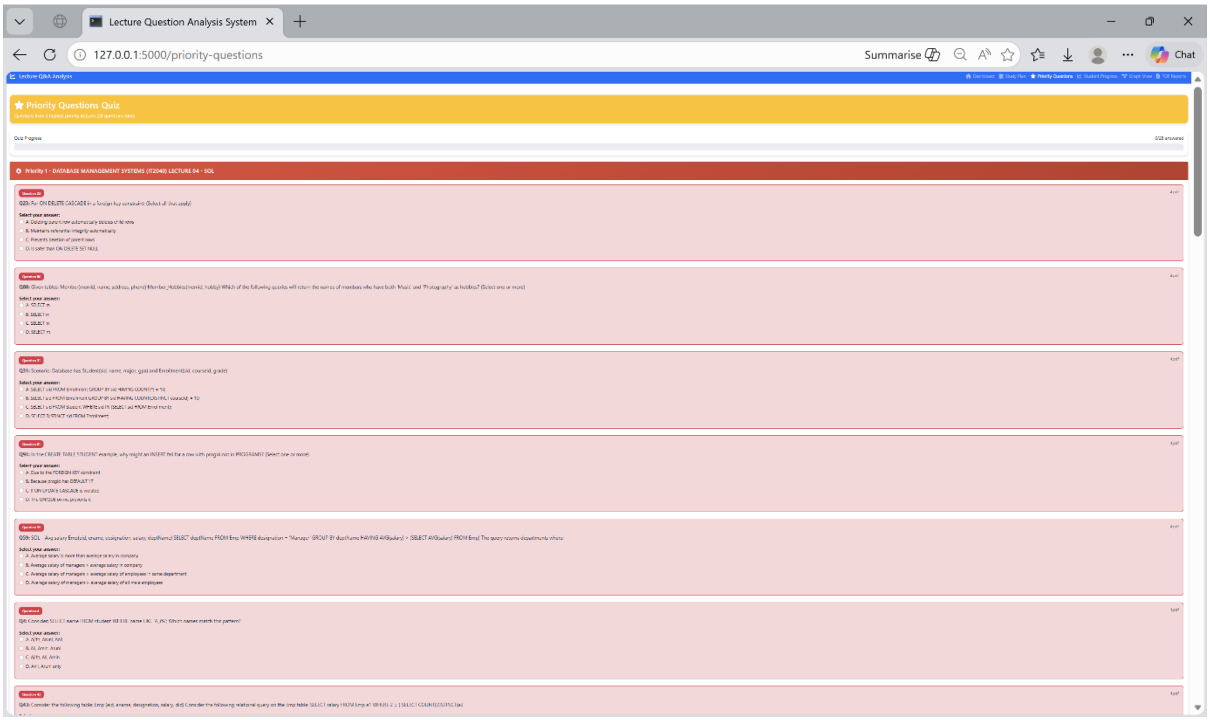
This page illustrates the generation of a baseline study plan based on exam question frequency. Using the lecture distribution analysis, the system allocates recommended study hours for each lecture according to its importance in past examinations. Lectures with higher question frequency receive greater study time allocation. The recommended focus areas table highlights priority lectures and suggested study hours for effective exam preparation.

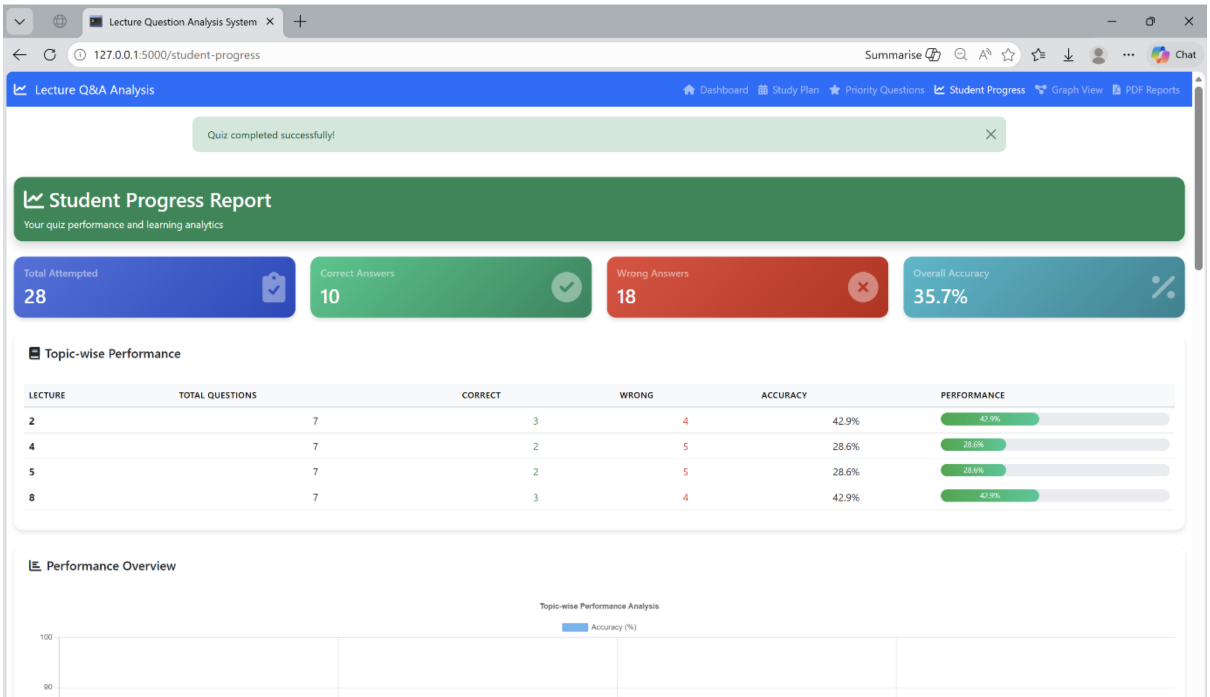


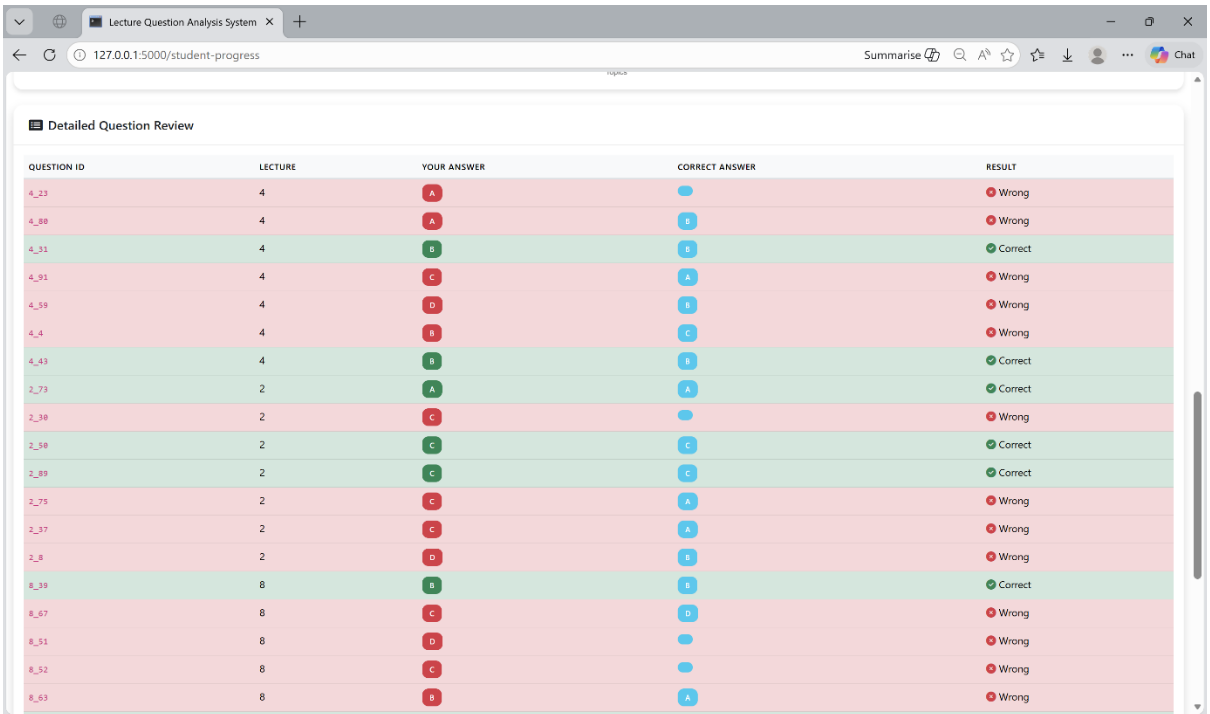


3. Student Quiz Evaluation and Progress Analysis

This page displays the student progress report after completing the recommended MCQ quiz. The system evaluates student responses and calculates performance metrics such as total questions attempted, number of correct answers, incorrect answers, and overall accuracy. Additionally, lecture-wise performance analysis helps identify topics where the student demonstrates strengths or weaknesses.

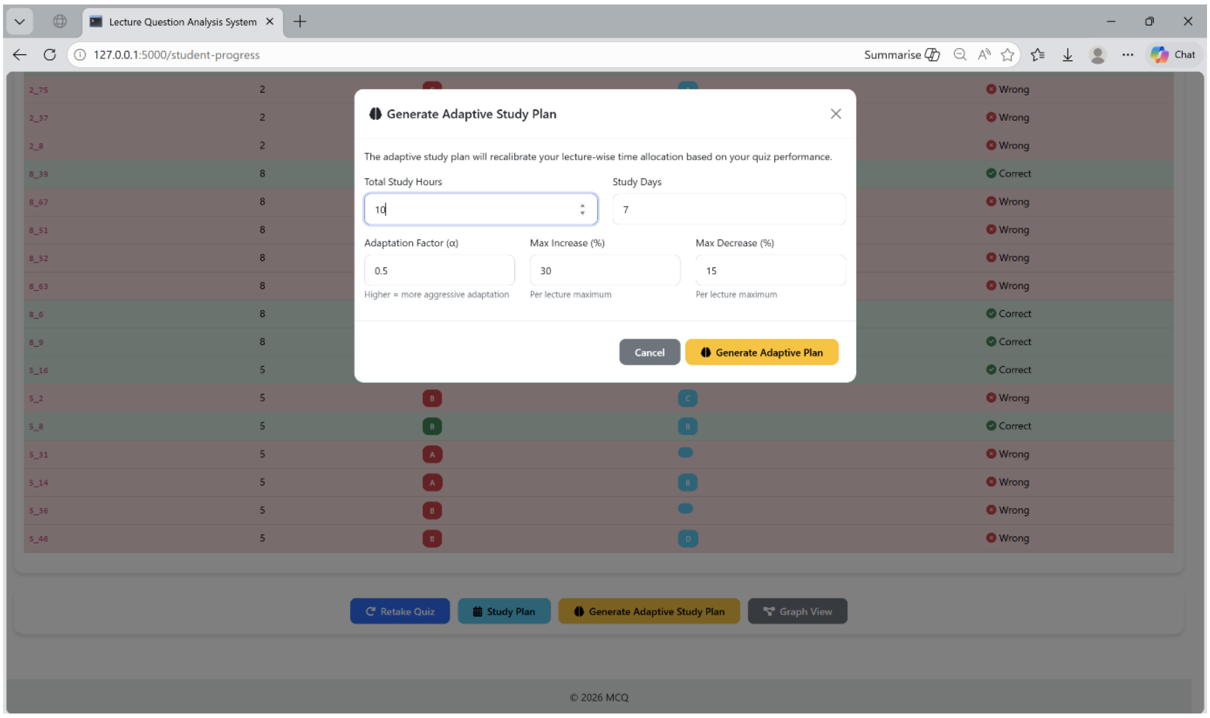


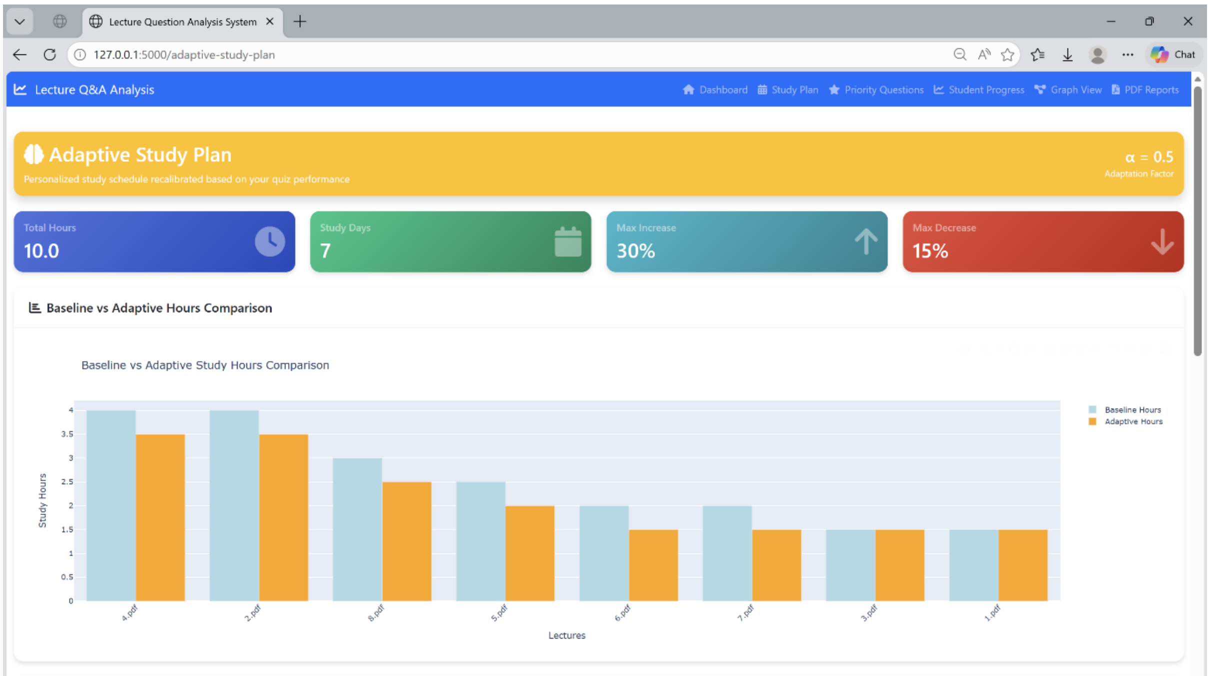




4. Adaptive Study Plan Generation

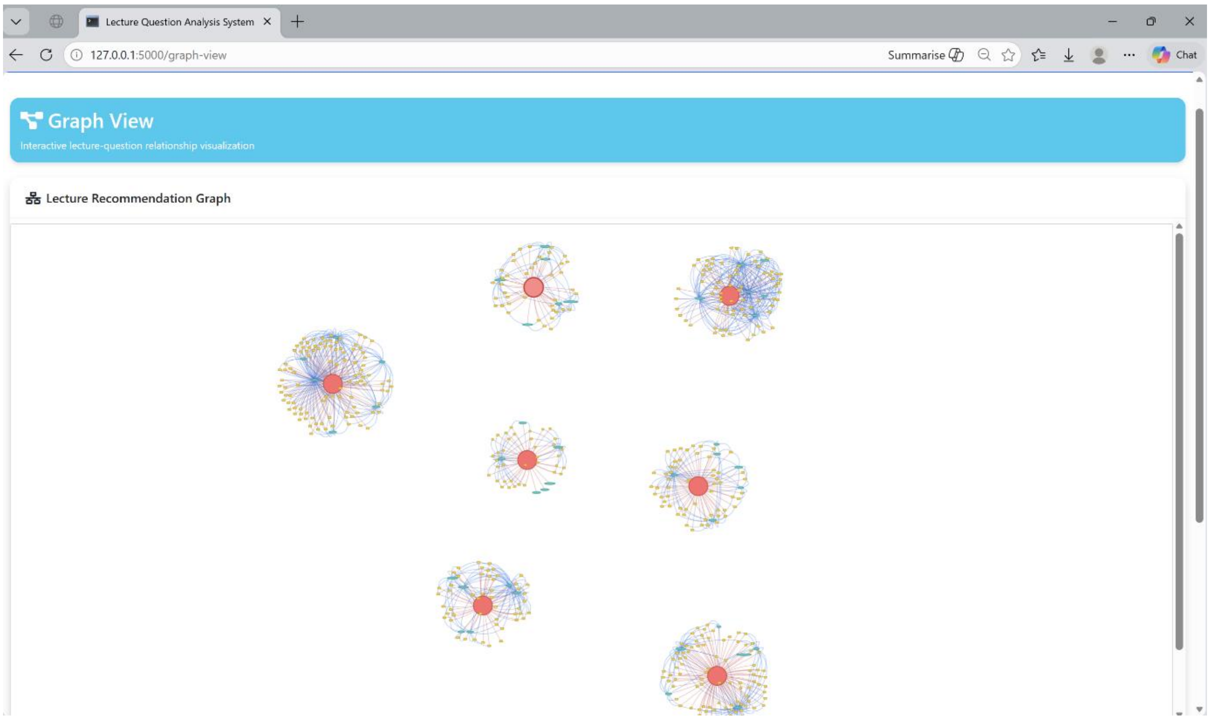
This page shows the adaptive study plan generation process. Based on the student's quiz performance, the system recalibrates the study plan by adjusting lecture-wise study hours. Weak lectures receive increased study time while strong lectures receive slightly reduced allocation. Controlled parameters such as adaptation factor and maximum adjustment limits ensure balanced modifications while maintaining a stable study schedule.

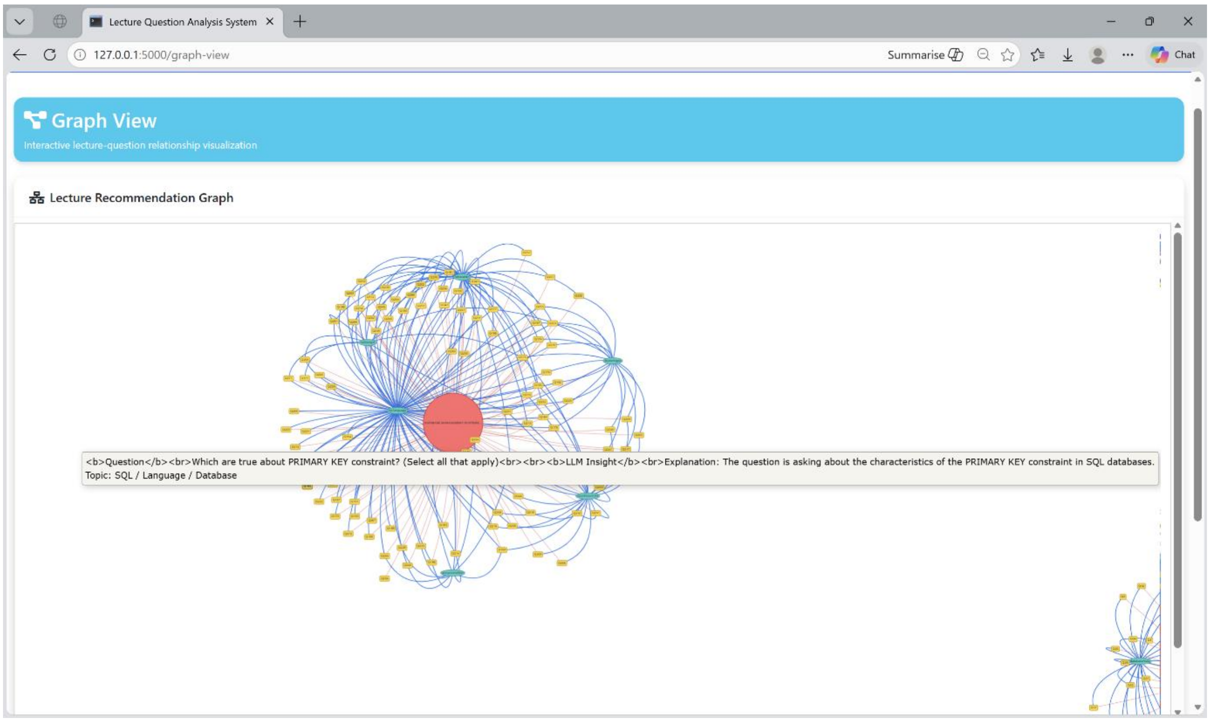




5. Graph-Based Visualization of Lecture–Topic–Question Relationships

This page presents a graph-based visualization of the relationships between lectures, topics, and MCQ questions. The graph structure is constructed using semantic similarity relationships identified through Natural Language Processing. This visualization enables exploration of how exam questions relate to lecture content and supports recommendation of relevant MCQs for targeted practice.





This is to confirm that the developed component functions in relation to the DMS module. The system analyzes DMS lecture slides and past paper MID MCQ questions using Natural Language Processing techniques. It performs semantic mapping between lecture topics and exam questions, conducts frequency-based analysis of past papers, and generates baseline and adaptive study plans to support exam preparation. The system also evaluates student quiz performance and applies a graph-based representation to model relationships between lectures, topics, and MCQ questions for visualization and recommendation.

DMS In-Charge:______________________________

Date:____________________